MA3521 — Separation of Variables

Name: Answer Key

Problem 1 $y' = e^{t+y^2}/y, \qquad y(1) = 2$

1a. Identify the factors in $y' = g(t)h(y)$.

$g(t) =$ e^t $\qquad h(y) =$ $\frac{e^{y^2}}{y}$

1b. Write the integrals. Do not evaluate.

$\int$ ye^{-y^2} $dy = \int$ e^t $dt + C$

1c. Evaluate the integrals.

$-\frac{1}{2}e^{-y^2}$ $=$ e^t $+ C$

1d. Evaluate the constant.

$C =$ $-\frac{1}{2}e^{-4} - e$

Problem 2 $y' = e^{t+y}, \qquad y(0) = 0$

2a. Identify the factors in $y' = g(t)h(y)$.

$g(t) =$ e^t $\qquad h(y) =$ e^y

2b. Write the integrals. Do not evaluate.

$\int$ e^{-y} $dy = \int$ e^t $dt + C$

2c. Evaluate the integrals.

$-e^{-y}$ $=$ e^t $+ C$

2d. Evaluate the constant.

$C =$ -2